Strategy D: A Non-σ-Complete, Non-Atomic, Measure-Free Boolean Algebra? — A Working Problem-Set

Patrick S. Mahon

Problem-set — June 2026

Abstract

A self-contained kit for attacking "Strategy D" and, concretely, its sharpest live entry point: *is the clopen algebra of the Argyros pre-Gleason space βY_n σ-complete?* Everything needed to sit with pen and paper is defined inline: Boolean algebras and Stone duality; σ-completeness, basic disconnectedness, cozero sets; measure-free algebras; the Argyros construction $(Y_n, \mathfrak{T})$ from [1]; and the Gillman–Jerison machinery [2] that reduces the question to one intrinsic predicate in Y_n. The open target is stated formally (§5) and broken into sub-tasks (§7) carrying the constraints established so far (three candidate families killed by one "interior-overshoot" mechanism; the strong-zero-dimensionality sub-lemma). Companion narrative + status: `strategy_d_dossier.md`, `argyros_sigma_completeness_handoff.md`. This document is presentation, not proof: the central question is open; included results are cited or short and self-verifying. GJ page scans: `notes/literature_review/literature/gillman_jerison_pages/`.

Contents

Why this matters

The question behind the question. This problem-set serves the *coherence/completion* programme: when an observer's system of finite distinctions is completed — pushed to its Stone space, the ideal limit of everything the distinctions can resolve — can that completed "horizon" always carry an honest (σ-additive) probability? In the classical/well-behaved case it can. Strategy D asks whether there is a completed horizon so structured that it *cannot*: a Boolean algebra (equivalently a Stone space) that is rich enough to be atomless and non-σ-complete, yet admits *no* strictly positive countably-additive probability at all. It is the sharp test of whether "coherence completes to probability" has genuine exceptions, or holds broadly.

What an answer decides (either way is informative):

- **Positive** (such a B exists) $\Rightarrow$ some coherent, non-atomic completed horizons are *structurally measure-free*: honest probability is not merely underdetermined there, it is *unavailable*. Probability-inhospitable horizons exist. (And a *ZFC* example — the live candidate here — would be a clean, axiom-free such object.)
- **Negative** (no such B) $\Rightarrow$ σ-additive probability is *broadly available* in the non-σ-complete non-atomic regime. The weight then shifts from *existence* to *selection*: not "can honest probability live here?" but "which admissibility/support condition — a CE-like commitment — picks out the relevant measure or governs descent to realised states?"

Why it is hard, and where it sits. The regime is genuinely delicate: the problem is likely independent of ZFC, and the nearest known constructions (Argyros, Kunen, Fedorchuk) each miss exactly one of the three conditions. The live entry point below is a single, ZFC, fully-specified candidate — the Argyros *pre-Gleason* space — where two of the three conditions hold outright and only σ-completeness is open. That makes it a rare thing: a deep set-theoretic-topology question with a completely concrete, hand-checkable reduction (§3–6). Whichever way it resolves, it sharpens the programme's central theme — whether the completion of a distinction system is automatically a probability space.

1 The problem, and the reductions that frame it

The open target (Strategy D). Does there exist a Boolean algebra B that is simultaneously

(i) *non-σ-complete*;
(ii) *non-atomic* (atomless); and
(iii) *measure-free*: B carries no strictly positive σ-additive probability measure?

(On a non-σ-complete B, "σ-additive" is read for disjoint countable families whose join exists in B.) Likely independent of ZFC; nearest known constructions miss at least one condition.

The three conditions translate, under Stone duality (§2), into three topological conditions on $K = \mathrm{St}(B)$; the live work is to find a K meeting all three. The sharpest current candidate is the *pre-Gleason* Argyros space (§4), where (ii),(iii) hold in ZFC and only (i) is open.

2 Boolean algebras and Stone duality

Definition 2.1 (Boolean algebra; the relevant notions)**.** A Boolean algebra B is a complemented distributive lattice $(B, \vee, \wedge, ', 0, 1)$.

- B is *σ-complete* if every countable subset has a supremum in B.
- An *atom* is a minimal nonzero element; B is *atomless* if it has none.
- A *measure* is a finitely additive $\mu\colon B \to [0, 1]$, $\mu(1) = 1$; *strictly positive* if $\mu(b) > 0$ for $b \neq 0$; *σ-additive* if $\mu(\bigvee_n b_n) = \sum_n \mu(b_n)$ for disjoint countable families with a join.

Theorem 2.2 (Stone duality)**.** $B \mapsto \mathrm{St}(B)$ *(the space of ultrafilters, with the topology generated by $\widehat{b} = \{U : b \in U\}$) is a duality between Boolean algebras and compact Hausdorff totally disconnected (*Stone*) spaces, with $B \cong \mathrm{Clop}(\mathrm{St}(B))$ (clopen algebra). Dictionary:*

Boolean B	***Stone space*** $K = \mathrm{St}(B)$
atomless	*no isolated points*
σ-complete	*basically disconnected (Def. 2.3)*
measure-free	*no strictly positive Radon probability measure*

Definition 2.3 (Cozero set; basically/extremally disconnected)**.** In a topological space X: a *cozero set* is $\mathrm{coz}(f) = \{x : f(x) \neq 0\}$ for continuous $f\colon X \to \mathbb{R}$ (in a zero-dimensional space, exactly the countable unions of clopens). Then [2, 1H]:

$$X \textit{ basically disconnected } :\Longleftrightarrow \ \mathrm{cl}(G) \text{ is open for every cozero set } G;$$

$$X \textit{ extremally disconnected } :\Longleftrightarrow \ \mathrm{cl}(U) \text{ is open for every open } U.$$

(Extremally $\Rightarrow$ basically.)

Theorem 2.4 (Loomis–Sikorski, Boolean form)**.** *For a Boolean algebra B: B is σ-complete $\Longleftrightarrow$ $\mathrm{St}(B)$ is basically disconnected $\Longleftrightarrow$ every countable subset $\{b_n\} \subseteq B$ has a supremum, realised by $\mathrm{cl}(\bigcup_n \widehat{b_n})$ being clopen.*

3 The Gillman–Jerison lever

Two facts from [2] (page scans in the repo) reduce "$\mathrm{St}(B)$ basically disconnected" to a condition on a base space and supply the witness shape.

Theorem 3.1 (βX transfer; GJ 6M.1)**.** *For a completely regular space X: βX (Stone–Čech) is basically disconnected $\Longleftrightarrow$ X is basically disconnected. (Likewise extremally; clean biconditional, no side condition.)*

Theorem 3.2 (The non-b.d. witness shape; GJ 6W)**.** *In $\beta\mathbb{N} \setminus \mathbb{N}$, the closure of the union of a* strictly increasing *sequence of clopen sets is never open; hence $\beta\mathbb{N} \setminus \mathbb{N}$ is not basically disconnected. More generally $\beta X \setminus X$ is not basically disconnected for infinite discrete X.*

Remark 3.3 (The route around βX — intrinsic, unconditional). For the question below the object is $B = \mathrm{Clop}(Y_n)$ and we need $\mathrm{St}(B)$ basically disconnected. By Thm. 2.4 this is *unconditional*: B is σ-complete $\Longleftrightarrow$ every countable clopen union in $\mathrm{St}(B)$ has clopen closure. Pulling back, this is intrinsic to $B = \mathrm{Clop}(Y_n)$ and needs no compactification. The GJ βX form (Thm. 3.1) gives the cleaner "compute in Y_n directly" route *only* under the sub-lemma that Y_n is *strongly* zero-dimensional ($\dim Y_n = 0$, equivalently βY_n zero-dimensional, equivalently $\mathrm{St}(\mathrm{Clop}\, Y_n) = \beta Y_n$); see Task 3. Thm. 3.2 gives the *witness template*: a strictly increasing clopen sequence with non-open-closure union.

4 The Argyros pre-Gleason construction $(Y_n, \mathfrak{T})$

From Argyros [1, §1.0–1.3], read directly. Fix $n \geq 2$.

Definition 4.1 (The tree T)**.** Choose pairwise-disjoint triples $S_{m,j} \in [\omega]^3$ ($m < \omega$, $1 \leq j \leq 3^m$). Set $T_m = \bigcup_j [S_{m,j}]^2$ (the 2-element subsets of each triple), so $|T_m| = 3^{m+1}$, and $T = \bigcup_m T_m \subseteq [\omega]^2$. Order: for $s \in T_m$, $t \in T_{m+1}$, $s \prec t$ iff $s, t \in [S_{m+1,j}]^2$ for some j. A *branch* Σ is a maximal chain; it selects one pair from a nested sequence of triples.

Definition 4.2 (Anti-diagonals and the space)**.** For a pair $s = \{k, l\} \in [\omega]^2$ let $K_s = \{x \in \{0,1\}^s : x_k \neq x_l\}$ (the *anti-diagonal*: the two assignments on s that differ). The space:

$Y_n = \{0,1\}^\omega$ as a set; $\quad \mathfrak{T} =$ topology with subbasis $\big($usual product-clopens$\big) \cup \{V_\Sigma : \Sigma \text{ a branch}\}$,

$$V_\Sigma = \prod_{s \in \Sigma} K_s \times \{0,1\}^{\omega \setminus \bigcup \Sigma} = \{x : x_k \neq x_l \text{ for every pair } \{k,l\} \in \Sigma\}.$$

Each V_Σ is *closed* in the usual topology, hence $\mathfrak{T}$-*clopen.* Basic $\mathfrak{T}$-clopen sets: $V = U \cap \bigcap_{j=1}^k V_{\Sigma_j}$, U usual-clopen. $(Y_n, \mathfrak{T})$ is completely regular Hausdorff, zero-dimensional (ind $= 0$: it has a clopen base), and has no isolated points.

Theorem 4.3 (Argyros 1.9 [1])**.** *For $n \geq 2$, $X_n := \beta Y_n$ has property $(*)$, hence the countable chain condition, and carries* no *strictly positive measure. (So* $\mathrm{Clop}(\beta Y_n)$ *is measure-free; and βY_n is compact, zero-dimensional, no isolated points.)*

Remark 4.4 (Pre-Gleason vs. Gleason — do not conflate). The famous "Argyros example" cited as *extremally* disconnected (Comfort–Negrepontis 6.25) is the *Gleason cover* $G(X_n)$, which is extremally disconnected (hence σ-complete — the *wrong* row for Strategy D) by construction. The present object is βY_n itself, *before* the Gleason step. Its σ-completeness is exactly what is open.

Fact 4.5 (Clopen algebras agree)**.** $\mathrm{Clop}(\beta Y_n) \cong \mathrm{Clop}(Y_n)$ *(each clopen of Y_n and its complement are completely separated, so extend uniquely to βY_n). So the question "*$\mathrm{Clop}(\beta Y_n)$ *σ-complete?" = "*$\mathrm{Clop}(Y_n)$ *σ-complete?"*

5 The open target

The open target (Step 1, the live entry point)*.* **Is** $\mathrm{Clop}(Y_n)$ **σ-complete?** Equivalently (Thm. 2.4, Fact 4.5, Def. 2.3):

Does every countable union of $\mathfrak{T}$-clopens in Y_n have $\mathfrak{T}$-open closure?

- **NO** (some countable clopen union has non-open closure) $\Rightarrow$ $\mathrm{Clop}(Y_n)$ non-σ-complete. Combined with Thm. 4.3 (measure-free) and no-isolated-points (atomless), βY_n is a **ZFC Strategy D example**.
- **YES** $\Rightarrow$ βY_n basically disconnected; this route is closed (fall back to a consistency construction under $\diamondsuit$/CH).

Remark 5.1 (Status). Open, with the prior leaning **toward YES (route closed)** — see the obstruction (§6). Not yet a proof either way.

6 What is established: three families killed by one mechanism

Three candidate non-σ-completeness families all turn out to *have* suprema; the third reveals a structural obstruction. (All elementary; verify by hand.) The relevant principle: in a field of sets, a countable family $\{A_k\}$ has a supremum $\iff$ $\mathrm{cl}(\bigcup_k A_k)$ is open (then it is the least clopen upper bound); the closure is the only obstacle.

Proposition 6.1 (Single branch)**.** *For one branch Σ, $V_\Sigma = \bigcap_m U^m$ for decreasing usual-clopens U^m; so $A_m = Y_n \setminus U^m$ increases to $Y_n \setminus V_\Sigma$, which is clopen (as V_Σ is). Set-union in the algebra $\Rightarrow$ sup exists.* $\times$

Proposition 6.2 (Coordinate / all-zeros)**.** *The point $\bar{0} = (0, 0, \dots)$ lies in* no V_Σ *(every pair-restriction of $\bar{0}$ is the diagonal $(0,0) \notin K_s$), so $\mathfrak{T}$ agrees with the usual topology at $\bar{0}$. With $C_m = [0^m 1]$ (usual-clopen), $\bigcup_m C_m = Y_n \setminus \{\bar{0}\}$, whose $\mathfrak{T}$-closure is Y_n ($\bar{0}$ a limit). Every clopen upper bound is Y_n;* $\sup = Y_n$ *exists.* $\times$

Proposition 6.3 (V_Σ-accumulation to a limit branch)**.** *Let branches $\Sigma_1, \Sigma_2, \dots$ share longer and longer initial segments, converging to a limit branch Σ_*; set $A_k = V_{\Sigma_k}$. Then $V_{\Sigma_*} \subseteq \mathrm{cl}_{\mathfrak{T}}(\bigcup_k V_{\Sigma_k})$ (an* overshoot*): post-divergence pairs of Σ_* and Σ_k lie on* disjoint *triples (Def. 4.1), so the anti-diagonal constraints are jointly satisfiable, so every $\mathfrak{T}$-neighbourhood of an $x \in V_{\Sigma_*}$ meets some V_{Σ_k}.* But this is not a witness: *V_{Σ_*} is itself a subbasic $\mathfrak{T}$-open set, so every overshoot point is* interior *to the closure; thus* $\mathrm{cl} = ($*open union*$) \cup ($*interior overshoot*$)$ *is open,* sup *exists.* $\times$

The obstruction. A witness needs a countable clopen union with *non-open* closure — a closure point that is *not interior.* But the new closure points produced by V_Σ-accumulation always land *inside* an open V_Σ, hence interior. So *V_Σ-built families cannot witness non-σ-completeness*: the same mechanism kills all three. A genuine witness needs a closure point lying in *no* V_Σ — a true boundary point — and the abundance of open V_Σ makes that hard. This shifts the prior toward "route closed." *(Strong structural argument, not yet a formalised theorem: the step "the only new closure points come from V_Σ-accumulation" wants tightening — see Task 2.)*

7 Sub-tasks — the pen-and-paper frontier

Witness side (lower-probability per the obstruction)

Sub-task 1 (A non-interior closure point)**.** Find a countable family of $\mathfrak{T}$-clopens $\{A_k\}$ and a point $x \in \mathrm{cl}_{\mathfrak{T}}(\bigcup_k A_k)$ such that x lies in *no* V_Σ contained in the closure — i.e. x has no $\mathfrak{T}$-open neighbourhood $\subseteq$ closure. (By §6, V_Σ-built families fail; this needs a genuinely different family. The all-zeros point $\bar{0}$ is the archetype of a "point in no V_Σ," yet its closure was still Y_n, open — understand why and whether any family avoids that.) Success $\Rightarrow$ ZFC Strategy D example.

Impossibility side (where the evidence points)

Sub-task 2 (Tighten the obstruction to a theorem)**.** Prove the structural claim of §6: *every* countable union of $\mathfrak{T}$-clopens has open $\mathfrak{T}$-closure (equivalently Y_n basically disconnected, equivalently $\mathrm{Clop}(Y_n)$ σ-complete). The crux is to show the only closure points of $\bigcup_k A_k$ beyond the (open) union itself arise from V_Σ-accumulation and are therefore interior. Use the basic-clopen normal form $V = U \cap \bigcap_j V_{\Sigma_j}$ (Def. 4.2) and Argyros's separation analysis [1, 1.3].

Sub-task 3 (The strong-zero-dimensionality sub-lemma)**.** Decide: is $(Y_n, \mathfrak{T})$ *strongly* zero-dimensional ($\dim Y_n = 0$)? This is *not* free from the clopen base ($\operatorname{ind} Y_n = 0$): $\mathfrak{T}$ is strictly finer than the product topology, which can break the Lindelöf condition that would upgrade $\operatorname{ind} = 0$ to $\dim = 0$. If **yes**, then βY_n is zero-dimensional, $\mathrm{St}(\mathrm{Clop}\, Y_n) = \beta Y_n$, and GJ 6M (Thm. 3.1) lets the whole question be computed as "every cozero set of Y_n has $\mathfrak{T}$-open closure" directly. Needed for the impossibility direction; the witness direction routes around it (Rmk. 3.3).

Sub-task 4 (Fallback: consistency construction)**.** If Y_n is basically disconnected (route closed), pivot to: construct a Strategy D algebra under $\Diamond$/CH (attack mode C of the dossier), and separately inspect the Kunen and Fedorchuk examples against the three conditions.

Remark 7.1 (Discipline). The basic-disconnectedness verdict is not Lean-checkable and the prior on a clean ZFC positive is low (decades of expert work nearby). Treat a candidate witness (Task 1) as a hypothesis to attack: is its closure point genuinely non-interior? Treat a candidate impossibility proof (Task 2) as needing the "only-V_Σ-accumulation" step rigorous, not assumed.

References

[1] S. A. Argyros, *On compact spaces without strictly positive measure*, Pacific J. Math. **105** (1983), no. 2, 257–272.

[2] L. Gillman and M. Jerison, *Rings of Continuous Functions*, Van Nostrand, 1960. (§1H basically/extremally disconnected; §6M βX transfer; §6W the increasing-clopen non-b.d. witness. Page scans: `notes/literature_review/literature/gillman_jerison_pages/`.)

[3] W. W. Comfort and S. Negrepontis, *Chain Conditions in Topology*, Cambridge Univ. Press, 1982. (Ch. 6; Thm. 6.25 = Gleason-cover Argyros example, Lemma 6.2 = Gleason spaces.)

[4] D. H. Fremlin, *Measure Theory*, vol. 5, §531–539 (measure-free / Maharam-type background).